

A Lean Approach to Estimate the Functional Size of Operating Applications

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Outline

- Motivation and proposal
- Empirical study
 - Design
 - Results and discussion
- Future work



Motivation

- Measuring the size of operating software applications
 - It is crucial for companies to have a clear “quantification” of the software applications stock
 - to support the decisions of the board of directors
 - to provide attestations with stakeholders
- How to measure them?
 - FSM methods: FPA approach widely adopted
- It is a complex task
 - applications developed in several past projects
 - evolved several times, developed by others, with long standing technologies, and without useful documentation



Motivation (2)

- It is often hard to find people having a complete view
 - Contact/interview several people to get information
 - Partial and inconsistent answers could be obtained
 - Undefined level of uncertainty in the measurement
 - Involve experts of the specific size measurement method
 - the measurement can be very expensive
- What can we do?
 - identify an approach to estimate the size using specific technical components, easily available to a project manager without employing further resources



Proposal

- Our proposal: use of number of physical tables and number of read and read/written tables
 - Obtained by exploiting the database schema and the application interfaces
 - This info is also employed by measurers to determine BFCs when a functional size measurement method is applied (e.g., IFPUG Function Points or COSMIC)
 - The idea motivating our research is similar to the one behind investigations proposing the use of single BFCs to provide estimates of software development effort in the early phases of software developmet



Research questions

- RQ1: *Can the number of physical tables be exploited to predict the application size (in terms of Function Points)?*
- RQ2: *Can the number of read and read/written tables be exploited to predict the application size (in terms of Function Points)?*



Data Set

- Data from InfoCamere: the ICT consortium company responsible for the design and management of the network of the Italian Chambers of Commerce nationwide
 - It has developed and manages the national system of data transmission connecting the 105 Chambers of Commerce and their 300 branch offices via a high speed, high security network
- The company designs, develops and delivers services via a Management System that complies with ISO 9001 (1997, 2000, 2008, 2009)



Data set (2)

- A set of 17 applications consisting of management information systems (50%) and applications mainly characterized by query and reporting systems (50%).
 - developed between 2005 and 2010, quite homogeneous in terms of adopted technologies (Java, Asp, PL/SQL), and development teams
- Number of (IFPUG) Function Points
 - in the software life cycle for each project of the company
 - contributes to the productivity database, together with the information on the development effort
 - internal benchmarking purposes and effort estimation



Data set (3)

- Number of physical tables, the number of read tables, and the number of read/written tables
 - Data collection by the same person of the company to improve consistency among measurements

Entity	Obs	Min	Max	Mean	Median	Std.Dev
FP	17	78	710	234.7	150	182.42
T	17	7	69	24.94	17	18.98
R	17	3	56	19.18	8	19.68
R/W	17	3	21	0.82	8	5.78



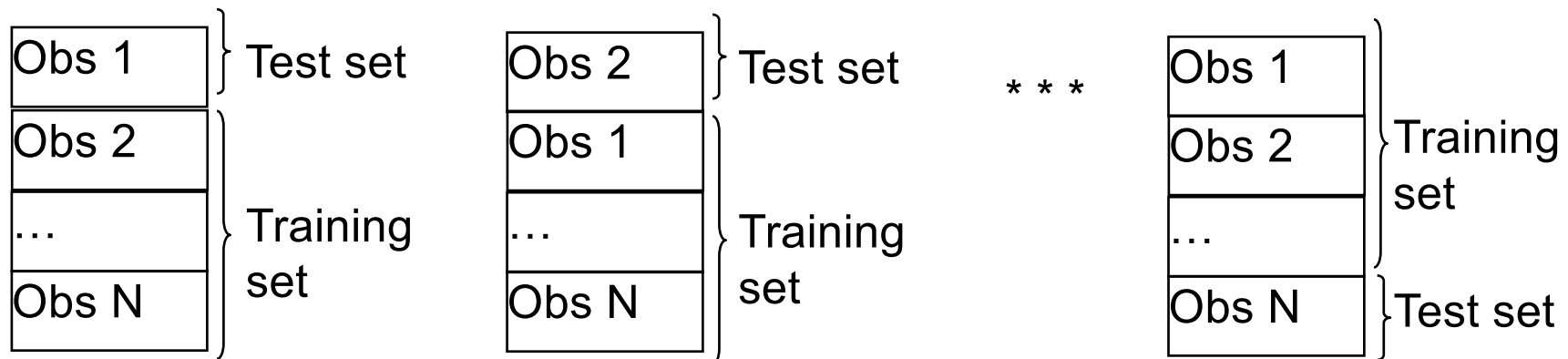
Estimation techniques

- (Linear Regression) $FP = b * T + c$
- (manual StepWise Regression) $FP = b_1 * R + b_2 * R/W + c$
 - At each step, the independent variable with the highest association (R^2) to the dependent variable, taking into account all the variables (i.e., R and R/W) currently in the model
 - To select variables that better explain the variation in the dependent variable
 - R^2 , shows the amount of the variance of the dependent variable explained by the model related to the independent variables
- Other indicators:
 - F-value and p-value for the confidence of the prediction
 - t-statistic of the variables for their significance



Validation method

- Leave-1-out cross validation
 - N different subsets of training (TS) and validation (VS) sets, where each validation set had one observation. TS used to build the estimation model with linear regression and VS used to assess the obtained estimate



Evaluation criteria

- MMRE & MdMRE: Mean & Median of Magnitude of Relative Error
 - $$\text{MRE} = |\text{Actual size} - \text{Predicted size}| / \text{Actual size}$$
 - An MMRE < 0.25 (i.e., 25%) acceptable (Conte et al.)
- $\text{Pred}(0.25) = k/N$
 - k is the number of obs with $\text{MRE} \leq 0.25$
 - A $\text{Pred}(0.25) > 0.75$ (i.e., 75%) acceptable (Conte et al.)
more than 75% of the estimated values fall within 25% of their actual values
- Boxplots of Absolute Residuals = $|\text{Actual size} - \text{Predicted size}|$
- Wilcoxon Test (a probability of 5% of committing a Type-I-Error)



Results for FP and T

- Application of LR
 - Normality of distributions not verified
 - Log transformation of FP and T
 - LR assumptions verified (Linearty, Homoscedasticity, Residual normality)
 - Analysis of outliers by residuals and Cook's distance
 - No influential observations

	Value	Std.Err	t-value	p-value
Intercept	2.715	0.238	11.39	<0.01
Coefficient of T	0.850	0.078	10.88	<0.01
	R ²	Std.Err	F	Sign F
	0.888	0.222	118.5	<0.01

$$\text{– LnFP} = 0.850 * \text{LnT} + 2.715 \longrightarrow \text{FP} = 15.105 * T^{0.850}$$



Results for FP and T (2)

- T is a good indicator of functional size
 - MMRE and MdMRE less than 0.25
 - Pred(0.25) quite close to 0.75

MMRE	MdMRE	Pred(0.25)
0.21	0.18	0.71

- The software company could obtain application size estimations just exploiting information on physical tables, with an error of about 21%

Results for FP and R and R/W

- Application of SWR
 - Normality of distributions not verified
 - Log transformation of FP and R and R/W
 - the best fitting model was the one including both R and R/W as independent variables
 - Influential observations were removed

	Value	Std.Err	t-value	p-value
Intercept	3.048	0.209	14.563	<0.01
Coefficient of Log R	0.486	0.048	10.563	<0.01
Coefficient of Log R/W	0.534	0.084	6.371	<0.01
	R ²	Std.Err	F	Sign F
	0.939	0.172	77.71	<0.01

- $FP = 21.073 * R^{0.486} * R/W^{0.534}$



Results for FP and R and R/W (2)

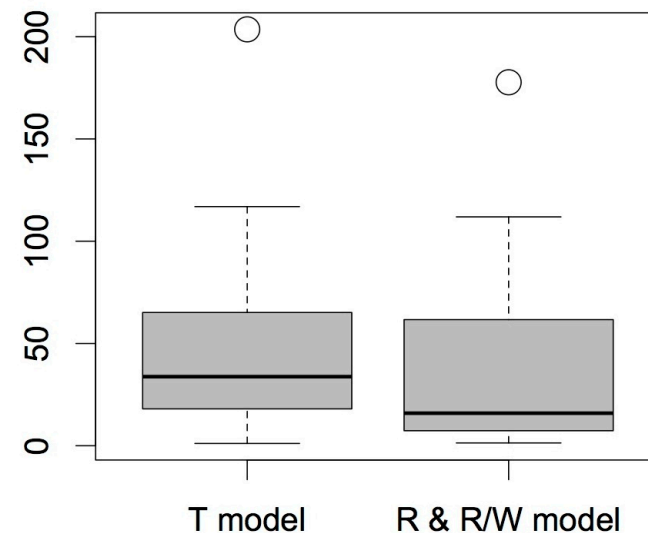
- R and R/W are good indicators of functional size
 - MMRE and MdMRE less than 0.25
 - Pred(0.25) quite close to 0.75

MMRE	MdMRE	Pred(0.25)
0.11	0.12	0.73

- The software company could obtain application size estimations just exploiting information on read tables and R/W tables, with an error of about 12%

Comparison and discussion

- We can positively answer our research questions:
 - The number of physical tables can be exploited to predict the application size in terms of FP
 - The number of read and read/written tables can be exploited to predict the application size in terms of FP
- R^2 and MMRE and MdMRE better for R and R/W
- No significant difference between the two models (wilcoxon test)



Comparison and discussion (2)

- The results interesting for InfoCamere
 - Relative errors considered acceptable
 - The asset of the applications stock can be estimated just looking at specific components easy to obtain
 - A cheap approach as support for decisions of the board of directors and for attestations with stakeholders

Comparison and discussion (3)

- Impact for InfoCamere
 - 4-6 hours to size a medium application by applying the traditional IFPUG Function Points Analysis (without considering the time spent to interview the people for searching the relevant information)
 - The use of tables required about 30 minutes
 - before applying the proposed method based on physical tables, InfoCamere tried to apply the LOC based backfiring approach
 - Considered not feasible by project managers
 - several programming languages, involvement of a developer of each application; complexity of the source code

Threats to validity



- Internal validity
 - reliability of the data and lack of standardization. Our best to control “extraneous factors”: use of the same questionnaires for all the applications and the project managers were instructed on how to use the questionnaires
- External validity
 - Replications in other industrial contexts



Threats to validity (2)



- Construct validity
 - The collection of the information to determine size measures represents a crucial aspect
- Conclusion validity
 - Evaluation criteria allowed assessing in an objective way the effectiveness of the predictions
 - We carefully applied the estimation methods and the statistical tests, verifying all the required assumptions
 - Low number of observations.

